Example Technological University

Faculty of Engineering

Programme: B.Tech Computer Science and Engineering

Course Code: CS501

Course Title: Machine Learning and Data Analytics

Semester: V Credits: 4

Academic Year: 2024-25

Course Objectives

1. To introduce the fundamentals of ML and supervised learning.

2. To develop data analysis skills using Python and pandas.

Unit I: Introduction to Machine Learning (8 Hours)
Types of learning; supervised learning and unsupervised learning. Bias-variance trade-off, cross-validation, overfitting. Linear algebra and statistics refresher.

Unit II: Data Preprocessing and SQL (8 Hours)
Data cleaning, feature engineering, handling missing values. Querying data with SQL and MySQ
Exploratory data analysis with pandas and NumPy.